Physics-Conditioned Video Diffusion with Kinematic Priors for Fusion Capsule Polishing

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Abstract

Video diffusion models can generate realistic process videos, but using them as controllable surrogates for industrial systems remains difficult when key control variables are only partially observable and no simulator or differentiable physics model is available. We study this regime in inertial confinement fusion (ICF) target capsule polishing, where surface defects can degrade fusion yield. The dynamics are governed by the polishing speed ω_p and capsule-pad slip S_C. While capsule motion can be monitored from video, S_C is not directly observable. To model these dynamics, we adapt a frozen Stable Video Diffusion (SVD-xt) backbone via LoRA, condition generation on Fourier encodings of (ω_p, S_C), and use an analytical kinematic model to guide training. Two training objectives operate on latent frame-difference dynamics: (i) a batch level correlation loss aligning them with the analytical sliding speed, and (ii) a stratified ratio-matching loss that, conditioned on ω_p, isolates slip structure from the dominant ω_p-driven variance. We evaluate kinematic consistency by tracking capsule motion in generated videos, inverting the kinematic model, and measuring rank agreement under a physical validity gate. For speed, rank accuracy exceeds 0.97 on real data; for slip, it reaches 0.735 with Spearman ≥ 0.50 on synthetic data. Overall, the results suggest a practical recipe for controllable industrial video surrogates when only low-dimensional analytical kinematics are available.

1 Introduction

Inertial confinement fusion (ICF) target capsules must be polished to nanometer tolerances because defects of only tens of nanometers can seed hydrodynamic instabilities and degrade ignition symmetry [30, 23, 2]. Polishing is therefore a costly throughput bottleneck [14, 7, 13], and improving it requires understanding how operating speed ω_p and capsule-pad slip S_C drive material removal. These dynamics are only partly observable: the capsule moves stochastically inside an opaque fixture [15], and routine instrumentation consists of a top-down video stream.

We seek *controllable video surrogates* that generate plausible videos of capsule motion from a real reference frame, a specified polishing speed ω_p, and a prescribed slip value S_C. These surrogates can support counterfactual sweeps over speed and slip, uncertainty estimates for motion quantities from repeated generation, and anomaly detection by comparing observed motion with plausible generated outcomes. Analytical polishing kinematics [26, 27] relate polishing speed and slip to capsule circulation, self-spin, and sliding speed. These relations guide training and provide a reference for assessing the kinematic consistency of generated motion. Most physics-guided video generation assumes PDE residuals, calibrated simulators, trajectory priors, dense motion supervision, or post-training guidance [5, 22, 47, 28, 51, 29, 42, 3, 45, 48]. Capsule polishing has none of these: the state is partly hidden, no calibrated video simulator exists for real videos, and the available physics is

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only a low-dimensional analytical map from (ω_p, S_C) to kinematic scalars. We therefore combine a stochastic video prior trained on real videos with scalar kinematic constraints.

Our approach adapts a frozen Stable Video Diffusion (SVD-xt) [6] image-to-video backbone with LoRA. Fourier tokens encode (ω_p, S_C) and enter cross-attention. The kinematics define two supervision signals without absolute labels: a batch correlation loss $\mathcal{L}_{\text{dyn}}$ aligning latent frame differences with analytical sliding speed, and a stratified ratio loss $\mathcal{L}_{\text{ratio}}$ matching within-ω_p latent dynamics ratios to the analytical slip ratio. Because real S_C is unobserved, we treat it as a prior-regularized latent conditioning variable and measure controllability by tracking generated clips and inverting them back to kinematics.

We combine these standard components to study controllable video generation under the sensing constraints of ICF capsule polishing. Our contributions are:

(i) Video generation with analytical kinematic supervision. We formulate a video surrogate that combines scalar conditioning with two latent objectives for analytical speed and slip relations. The formulation uses process videos and analytical kinematics without requiring a calibrated simulator of the real process or dense motion labels.

(ii) Kinematic evaluation of generated motion. We recover capsule circulation from generated pixels and use kinematic inversion to assess speed and slip ordering. We report rank accuracy, confidence intervals, and the fraction of clips passing validity checks alongside visual metrics. These scores measure consistency with the analytical model shared by training and evaluation.

(iii) Speed control and the limits of slip recovery. We distinguish observable speed control in real videos from slip ordering in synthetic videos with known labels. Across five evaluation seeds, the full model reaches real speed rank 0.985 and synthetic slip rank 0.735 (Tables 1 and 2). Real slip ordering remains at chance. Ablations show that scalar conditioning supplies most of the ordering, while the combined losses reduce synthetic FID-I3D and FVD without improving slip rank over conditioning alone.

2 Related Work

Video diffusion and scalar conditioning. Latent and video diffusion models provide strong short-horizon visual priors [33, 20, 6, 31], with LoRA enabling adaptation to small industrial datasets [21]. Continuous process controls require mechanisms that distinguish nearby physical values; common choices include FiLM, cross attention, ControlNet or IP-Adapter branches, and Fourier features [32, 49, 46, 36].

Physics guided video generation. Recent methods inject physics as field residuals or score guidance for known PDEs [5, 22], simulators for rigid bodies, materials, or interactions [47, 28, 8, 51, 42, 3], or dense optical-flow/trajectory signals [29, 43, 53]. Object-centric methods similarly assume spatial descriptors such as masks, contact fields, or skeleton tracks. Capsule polishing lacks these signals: the state is hidden, dense correspondences are unreliable on a featureless specular shell, and the only physics is the scalar pair (ω_p, S_C). We compare against flow and analytical warp baselines in physical coordinates (Sec. 6.5); dense-flow or trajectory-token baselines require supervision that is unavailable here. On this same apparatus, supervised models predict roughness and polish rate from process signals [38, 12], a public video and sensor dataset supports method development [4], and a companion study uses generative modeling for multi-modal anomaly scoring rather than counterfactual video synthesis [1].

LoRA prediction and hypernetwork control. Recent controllers can alter model weights or use scaling tokens: TC-LoRA predicts time-varying LoRA weights, Video2LoRA predicts LoRA modules from a reference video [9, 44], and LiON-LoRA-style fusion designs use tokens to modulate motion strength. Our controls are physical scalars supervised by kinematic ordering, so we keep one LoRA adapter set and inject Fourier operating tokens at every cross-attention layer. This gives a simple conditioning-only baseline analogous to an inference-time scaling-token controller: the "conditioning only" rows keep the tokens but remove $\mathcal{L}_{\text{dyn}}, \mathcal{L}_{\text{ratio}}$, isolating the marginal effect of the losses.

Kinematic-consistency evaluation. General video benchmarks for physical plausibility and scientific diffusion emulators emphasize that perceptual realism is not enough for physical use. Our metric follows that principle at the domain scale: instead of reporting only FID/FVD, we recover $\widehat{\omega_c}$ and $\widehat{R_T}$

from generated pixels and score monotonic agreement with operating variables. This does not make the videos mechanically exact, but it prevents appearance metrics from being mistaken for process controllability.

3 Preliminaries

Problem. Given a dataset $\mathcal{D} = \{(\mathbf{v}_i, \omega_p^{(i)}, S_C^{(i)}, \mathbf{x}_{\text{cond}}^{(i)})\}_{i=1}^N$ of clips $\mathbf{v}_i \in \mathbb{R}^{T\times 3\times H\times W}$, baseplate speed $\omega_p^{(i)}$, head-contact slip $S_C^{(i)}$, and first frame $\mathbf{x}_{\text{cond}}^{(i)}$, we learn $p_\theta(\mathbf{v} \mid \omega_p, S_C, \mathbf{x}_{\text{cond}})$ which clips are visually realistic and whose dynamics scale with $V_{\text{eff}}(\omega_p, S_C)$ (speed) and $R_T(S_C)$ (slip).

Analytical kinematics.

Following Lee's kinematic model [26, 27], a rigid ball of radius R_b rests in a V-groove of half-angle θ and radius R_g on a baseplate rotating at ω_p. No slip at the two groove contacts and head-contact slip $S_C \in [0, 1]$ give the scalar map below (derivation in Appendix A1); Fig. 1 shows the contacts.
Two facts drive the losses: for fixed slip, V_{eff} is linear in ω_p with zero intercept; for $\theta \in (0, \pi/2)$, R_T is strictly increasing in S_C and provides an ordinal slip target.

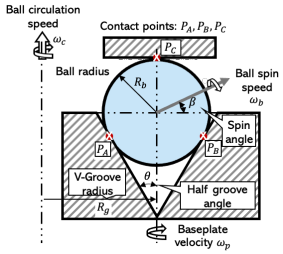


Figure 1: Polishing setup.

$$\omega_c = \omega_p \frac{1 + S_C \sin\theta}{1 + \sin\theta}, \quad \omega_b = \frac{R_g\, \omega_p\, (1 - S_C)}{R_b \cos\beta\, (1 + \sin\theta)}, \quad V_C = S_C\, R_g\, \omega_p, \tag{1}$$

where ω_c is shell circulation, ω_b is self-spin, V_C is head-contact sliding speed, and $\cos\beta = R_g/\sqrt{R_g^2 + R_b^2 \sin^2\theta}$ it is set by geometry. We define

$$V_{\text{eff}}(\omega_p, S_C) = \sqrt{(R_g\omega_c)^2 + (R_b\omega_b)^2 + V_C^2}, \qquad R_T(S_C) \triangleq \frac{\omega_c}{\omega_p} = \frac{1 + S_C \sin\theta}{1 + \sin\theta}. \tag{2}$$

Our apparatus has θ=45° and speeds $\{21, 48, 78, 108\}$ RPM. Real clips use shells bonded to the baseplate, so the head contact is nominally pure sliding (S_C=1) [26]; compliant-pad deformation, pressure gradients, and fluid effects can lower the effective slip [24], but per-clip S_C is unobservable from overhead video. We therefore use $S_C \sim \text{Beta}(199.5, 10.5)$ (mean 0.95), rescaled to $[0.92, 0.98]$, as an uncertainty prior and regularizer, not a measured label [16]. To test slip when labels exist, we also build synthetic videos with $S_C \in \{0.20, 0.40, 0.60, 0.80\}$ (Sec. 6).

4 Method

The framework has three parts: a frozen video backbone with LoRA adapters, Fourier physics tokens injected by cross attention, and latent losses computed from differentiable kinematics.

Backbone. We adopt SVD-xt [6], which operates on VAE latents $\mathbf{z} \in \mathbb{R}^{T\times C\times h\times w}$ with v-prediction $\mathbf{v}_\theta = \sqrt{\bar\alpha_t}\boldsymbol{\epsilon} - \sqrt{1-\bar\alpha_t}\,\mathbf{x}_0$ and clean-latent recovery $\hat{\mathbf{x}}_0 = \sqrt{\bar\alpha_t}\,\mathbf{z}_t - \sqrt{1-\bar\alpha_t}\,\mathbf{v}_\theta$. We freeze the backbone, VAE, and CLIP encoder, and train rank-32 LoRA adapters [21] on attention projections with Min-SNR-γ weighting (γ=5) [17].

Fourier physics tokens. RPM is normalized to $\widehat{\omega_p} = (\omega_p - \omega_{p,\min})/(\omega_{p,\max} - \omega_{p,\min}) \in [0, 1]$ over $\{21, 48, 78, 108\}$ RPM; S_C is already fractional. Each scalar s is lifted to $\gamma(s) = \big[\sin(2^k\pi s), \cos(2^k\pi s)\big]_{k=0}^{K-1}$ and encoded by an MLP into $\mathbb{R}^d$ (d=1024). The continuity of γ supports interpolation to unseen speeds, tested by leave-one-speed-out evaluation (Sec. 6); it also permits off-grid queries such as 55 RPM, though we only validate held-out grid speeds here. Tokens $\mathbf{T}_{\text{rpm}}, \mathbf{T}_{S_C}$ are combined by multi-head attention with the mean-token query, and the conditioning sequence is

$$\mathbf{T}_{\text{int}} = \text{LN}\big(\bar{\mathbf{T}} + \text{MHA}(\bar{\mathbf{T}}, \mathbf{T}_{12}, \mathbf{T}_{12})\big), \quad \mathbf{E} = [\mathbf{e}_{\text{img}}; \mathbf{T}_{\text{rpm}}; \mathbf{T}_{S_C}; \mathbf{T}_{\text{int}}] \in \mathbb{R}^{4\times d}, \tag{3}$$

with $\mathbf{T}_{12} = [\mathbf{T}_{\text{rpm}}; \mathbf{T}_{S_C}]$ and $\bar{\mathbf{T}} = \frac{1}{2}(\mathbf{T}_{\text{rpm}} + \mathbf{T}_{S_C})$. $\mathbf{E}$ enters every UNet cross-attention layer through LoRA-adapted K, V projections; Q stays frozen to reduce query drift. Self-attention Q, K, V and output projections also receive LoRA adapters (r=α=32) across temporal and spatial

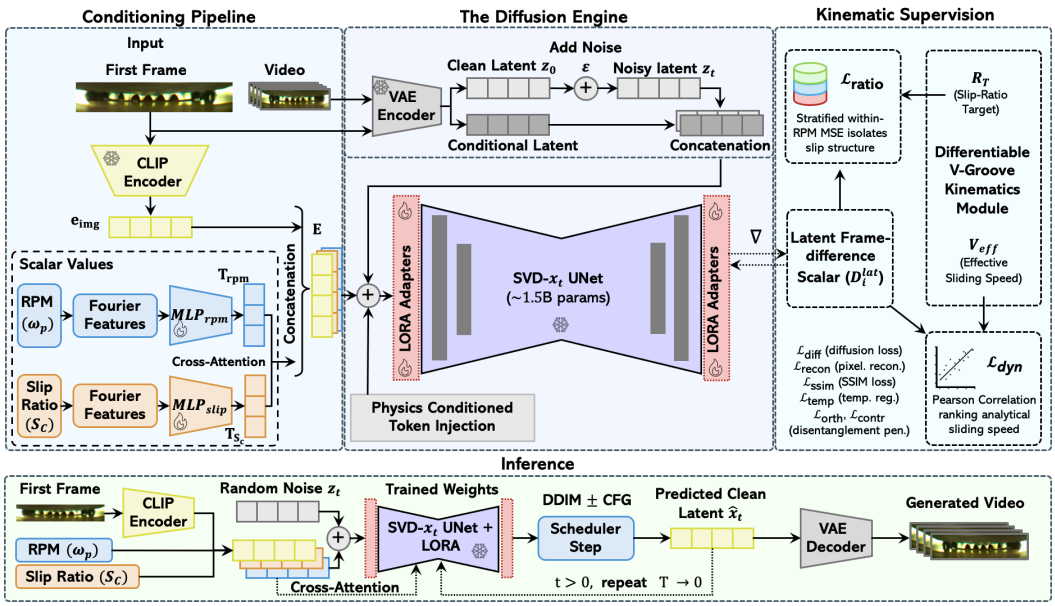


Figure 2: Method overview. SVD-xt is frozen (*gray*); only LoRA adapters and the Fourier physics encoder (*blue*) are trained. The operating scalars (ω_p, S_C) become cross-attention tokens and kinematic targets V_{eff}, R_T for latent losses $\mathcal{L}_{\text{dyn}}, \mathcal{L}_{\text{ratio}}$ on $\hat{\mathbf{x}}_0$. At inference (bottom row) there is no target clip: starting from random noise, the same frozen encoders and trained adapters are reused to alternate the UNet with the sampling scheduler, and the frozen VAE decodes only the final latent to pixels.

attention. Image and physics tokens are jointly dropped with p_{cfg}=0.1 for classifier-free guidance [19]. Encoding and mechanism ablations are in Sec. 6: FiLM replaces cross attention with a feature-wise affine modulation of the UNet activations, and raw S_C keeps cross attention but removes the Fourier lift γ for S_C, feeding the scalar directly.

Differentiable kinematics module. Equations (1)–(2) map physical (ω_p, S_C) to $V_{\text{eff}}, \omega_c, \omega_b, R_T$. These are scalar targets only: gradients enter LoRA through the latent frame-difference scalar D^{lat} computed on $\hat{\mathbf{x}}_0$, not through the kinematic constants. Kinematics use physical RPM; only tokens use normalized RPM.

Dynamics consistency loss $\mathcal{L}_{\text{dyn}}$. From $\hat{\mathbf{x}}_0$ we form a per-clip latent frame-difference scalar

$$D_i^{\text{lat}} = \frac{1}{(T-1)Chw} \sum_{t=1}^{T-1} \left\| \hat{\mathbf{x}}_{0,i}^{(t+1)} - \hat{\mathbf{x}}_{0,i}^{(t)} \right\|_1, \qquad \mathcal{L}_{\text{dyn}} = -\operatorname{corr}\left(V_{\text{eff}}^{(1:B)}, D^{\text{lat}\,(1:B)}\right). \tag{4}$$

Unlike PDE-residual losses [5] that constrain each sample, $\mathcal{L}_{\text{dyn}}$ enforces a *ranking* across samples consistent with the analytical sliding speed.

Stratified velocity-ratio loss $\mathcal{L}_{\text{ratio}}$. Naively replacing V_{eff} with $R_T(S_C)$ in Eq. (4) fails because between-RPM dynamics dominate, while the within-RPM variance of $R_T(S_C)$ under the real Beta prior is only $\operatorname{Var}(R_T \mid r) \approx 3.85 \times 10^{-5}$. A pooled correlation, therefore, sees mostly RPM. so we stratify by RPM. This also explains why the loss is a weak signal at B≈8–16: it regularizes ordinal structure rather than providing dense per-sample supervision. For each stratum $\mathcal{B}_r$ in the batch, define mean-normalized signals $\tilde{D}^{(i)} = D_i^{\text{lat}} / \text{sg}\left[\overline{D^{\text{lat}}}_{\mathcal{B}_r}\right]$, $\tilde{R_T}^{(i)} = R_T(S_C^{(i)}) / \overline{R_T}_{\mathcal{B}_r}$, where $\overline{D^{\text{lat}}}_{\mathcal{B}_r} = \frac{1}{|\mathcal{B}_r|} \sum_{j \in \mathcal{B}_r} D_j^{\text{lat}}$ and analogously for $\overline{R_T}_{\mathcal{B}_r}$. The stop-gradient $\text{sg}[\cdot]$ (`.detach()`) on the stratum-mean denominator prevents the loss from collapsing the predicted ratios to a degenerate scale. We use

$$\mathcal{L}_{\text{ratio}} = \frac{1}{|\mathcal{R}_B|} \sum_{r \in \mathcal{R}_B} \frac{1}{|\mathcal{B}_r|} \sum_{i \in \mathcal{B}_r} \left(\tilde{D}^{(i)} - \tilde{R_T}^{(i)}\right)^2, \tag{5}$$

including only strata with $|\mathcal{B}_r| \geq 2$ and within-stratum $\mathrm{Var}(R_T) > \varepsilon$. Mean-normalization removes RPM scale; MSE keeps a smooth gradient at $|\mathcal{B}_r|=2$, where Pearson is degenerate at ± 1.

Role of the Beta prior on S_C. In the real regime, S_C is not measured per clip. We sample $S_C \sim \mathrm{Beta}(199.5, 10.5)$ rescaled to $[0.92, 0.98]$ and use the sample as the conditioning token input and as the scalar argument to the kinematic targets (V_{eff}, R_T). The prior has two limited roles: it keeps the S_C token in a plausible range and supplies weak within-stratum order for $\mathcal{L}_{\text{ratio}}$. Both physics losses are ranking objectives, not absolute slip regression; even a ± 0.03 slip perturbation changes R_T by only about 1.2 percentage points at $\theta=45^\circ$. The Beta prior is therefore a regularizer, not a supervision oracle, and real-slip claims do not rely on it as ground truth. A wider prior would strengthen $\mathcal{L}_{\text{ratio}}$ gradients but would feed real clips slip values unsupported by the apparatus; a narrower prior would be physically safer but would nearly remove the slip signal. We choose the conservative middle and evaluate true slip only in S.

Auxiliary objectives and a two-phase schedule. A latent reconstruction loss $\mathcal{L}_{\text{recon}} = \ell_1(\hat{\mathbf{x}}_0, \mathbf{x}_0)$, $\mathcal{L}_{\text{SSIM}} = 1 - \mathrm{SSIM}(\hat{\mathbf{x}}_0, \mathbf{x}_0)$, $\mathcal{L}_{\text{temp}}$, and small token penalties $\mathcal{L}_{\text{orth}}, \mathcal{L}_{\text{contr}}$ (Appendix A4.2) complement $\mathcal{L}_{\text{diff}}$. Because physics losses need coherent $\hat{\mathbf{x}}_0$, Phase 1 trains perceptual losses only, and Phase 2 ramps physics terms with $w(e)$.

$$\mathcal{L} = \mathcal{L}_{\text{diff}} + \lambda_r \mathcal{L}_{\text{recon}} + \lambda_s \mathcal{L}_{\text{SSIM}} + w(e)\big[\lambda_d \mathcal{L}_{\text{dyn}} + \lambda_t \mathcal{L}_{\text{temp}} + \lambda_\rho \mathcal{L}_{\text{ratio}} + \lambda_o \mathcal{L}_{\text{orth}} + \lambda_c \mathcal{L}_{\text{contr}}\big]. \quad (6)$$

Defaults are reported in Appendix A4.

Inference. At test time, 25-step DDIM [35] with guidance scale $s=3.0$ generates a $T=30$-frame clip from a first frame and target $(\omega_p^\star, S_C^\star)$.

5 Theoretical Insights

The losses use only scalar kinematics, but their batch form needs three checks: $\mathcal{L}_{\text{dyn}}$ must be a valid correlation objective, $\mathcal{L}_{\text{ratio}}$ must expose the weak slip axis, and D^{lat} must preserve visible-motion ordering. Full derivations are in Appendix A2.

5.1 Batch reliability of $\mathcal{L}_{\text{dyn}}$

$\mathcal{L}_{\text{dyn}}$ is the negative batch Pearson correlation between $V_{\text{eff}}(\omega_p, S_C)$ and D^{lat} (Eq. 4). It is well posed when both have nonzero batch variance. RPM-stratified sampling supplies distinct speeds, and the two-phase schedule waits until reconstructed latents have temporal motion.

Proposition 1 (Well-posed batch correlation)**.** *For $B \geq 2$, if $s^2_{B,V_{\text{eff}}} > 0$ and $s^2_{B,D^{\text{lat}}} > 0$, the batch Pearson correlation in $\mathcal{L}_{\text{dyn}}$ is defined, bounded in $[-1, 1]$, locally Lipschitz in the denoiser parameters away from the zero-variance set, and differentiable almost everywhere. For $B > 3$, under a bivariate normal reference model with population correlation ρ,*

$$\sqrt{B-3}\,\big(z_F(\hat{r}_B) - z_F(\rho)\big) \xrightarrow{d} \mathcal{N}(0,1), \quad z_F(r) = \tfrac{1}{2}\log\tfrac{1+r}{1-r}. \quad (7)$$

Thus, under this asymptotic reference approximation, a Fisher-z half-width ε at confidence $1-\delta$ corresponds to $B \approx 3 + z^2_{1-\delta/2}/\varepsilon^2$.

At $\varepsilon=0.1, \delta=0.05$, $B^*=388$. Our effective batches (8 on most real runs, 16 on synthetic LOSO) therefore use $\mathcal{L}_{\text{dyn}}$ as a noisy ordering signal, not a precise population estimate. Figure 3(a,b) shows the Fisher uncertainty and nonzero denominators under stratified batching.

5.2 Why $\mathcal{L}_{\text{ratio}}$ is stratified

$R_T(S_C)$ is affine in S_C with slope $m = \sin\theta/(1+\sin\theta) \approx 0.414$ at $\theta=45^\circ$. Under the real Beta prior, $\mathrm{Var}(R_T \mid r) \approx 3.86 \times 10^{-5}$ is within an RPM.tratum, whereas V_{eff} changes by about $5\times$ from 21 to 108 RPM. A pooled slip correlation is therefore speed-dominated. Eq. (5) removes the RPM component, mean-normalizes strata, and uses MSE, so two-sample strata still have smooth gradients.

5.3 Latent and pixel motion

The physics losses use D^{lat} instead of optical flow or pixel differences. On 120 real clips (Fig. 3(c)), D^{lat} has pooled Pearson $r=0.91$ with pixel frame difference, per-RPM correlations

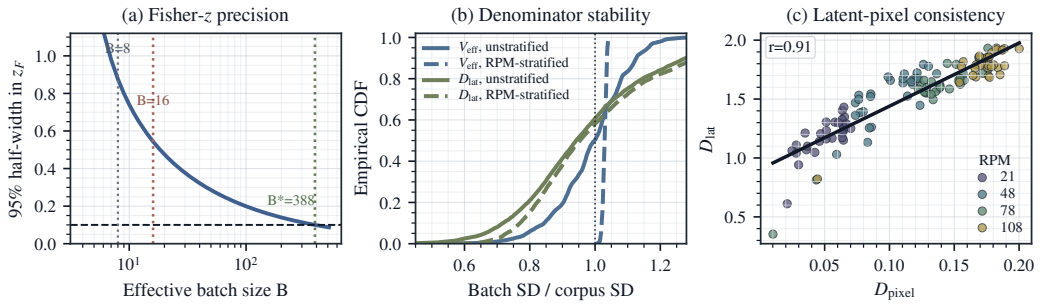


Figure 3: Foundations of the physics losses. **(a)** Fisher-z uncertainty shows that $\mathcal{L}_{\text{dyn}}$ is a noisy batch signal at our compute budget. **(b)** RPM-stratified batching keeps the $\mathcal{L}_{\text{dyn}}$ denominator away from collapse. **(c)** The latent motion scalar D^{lat} tracks pixel motion on real clips at $r{=}0.91$.

$\{0.78, 0.83, 0.94, 0.92\}$, and low-band/high-band correlations $0.985/0.730$. Thus, it captures circulation-like visible motion more strongly than self-rotation; we treat this as a scope limit, not a contact-mechanics guarantee.

Scope. These arguments support scalar ordering objectives only. Generated videos are not claimed to reproduce contact mechanics, material removal, or roughness evolution; the training signal is a cross-clip kinematic prior, not a pixel-level physical law.

6 Experiments

$S_C \in [0, 1]$ denotes head-contact slip, from pure rolling (0) to pure sliding (1) [26]. Because real S_C is unmeasured, regime R evaluates realism and speed control under a narrow prior, while synthetic regime S tests slip recovery with known S_C.

6.1 Datasets and evaluation

Real regime R**.** Four videos at $\{21, 48, 78, 108\}$ RPM yield 604 clips of $T{=}30$ frames (stride 5), split $70/20/10$ by source segment so overlapping windows do not cross splits. Each clip receives a persisted draw $S_C \sim \text{Beta}(199.5, 10.5)$ over $[0.92, 0.98]$ (Sec. 3); this regime grounds speed and realism but has no usable slip label.

Synthetic regime S**.** A renderer driven by Eqs. (1)–(2) produces matched videos at the same RPMs and known slips $\{0.20, 0.40, 0.60, 0.80\}$, supplying the wide slip variation absent from R. It is a positive-control renderer rather than a claim of photorealistic polishing: it matches the orbit geometry and timing used by the scorer but only approximates real lighting, fluid, specularities, and partial occlusions. We therefore use S to test identifiability under known labels, not to claim synthetic-to-real slip transfer. Because renderer and scorer share kinematic geometry, success on S is best read as a controlled observability test with known slip labels, not an independent physics audit; the real-domain conclusion still comes from R under the current evaluation setup.

Metrics. Visual metrics are paired PSNR, SSIM, LPIPS, FID-I3D, and FVD [18, 40]; FID-I3D uses the FVD RGB-I3D Kinetics-400 backbone, so comparisons are relative. Control uses a $4 \times 4 \times 4$ sweep per seed (scene $\times$ RPM $\times$ slip): speed rank orders $\widehat{\omega_c}$ across RPMs at fixed scene/slip, and slip rank orders $\widehat{R_T}$ across slips at fixed scene/RPM. Real sweeps use $S_C \in \{0.92, 0.94, 0.96, 0.98\}$; synthetic sweeps use the labeled S grid. Results average five seeds unless a deterministic baseline is identified.

Tracking and inversion. Scoring uses fine-tuned YOLO11n detections stitched into tracks, circulation recovery $\widehat{\omega_c}$, and Eq. (1) inversion (Fig. 4). It is independent of training: losses use latent D^{lat} and scalar $V_{\text{eff}}(\omega_p, S_C)$, while scoring uses only pixels, tracking gates (length, singularity, R^2, cross-track MAD, sign, aggregation, plausibility), and a validation-fitted calibration c_{suite} applied before inversion. This scalar is fit once per suite on calibration clips and then fixed across variants, seeds, and test sweeps. No test labels, D^{lat}, or loss values enter scoring. Yield is the fraction of plausible sweep cells; ranks or correlations use passing cells only and are reported with yield where

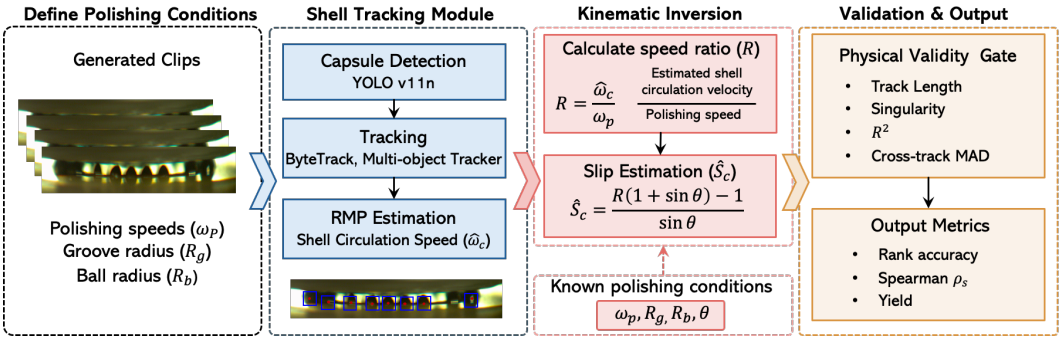


Figure 4: Slip evaluation. Generated clips pass through capsule detection, tracking, circulation recovery, kinematic inversion, and a physical validity gate before rank accuracy is computed.

Table 1: Real design study (R), five-seed mean. Visual metrics use the paired test split; speed/slip use the gated sweep. Chance slip rank is 0.5; bold marks best within group; † marks default. Yield is the fraction of clips passing the physical validity gate.

		Visual					Speed		Slip		
Group	**Variant**	PSNR↑	SSIM↑	LPIPS↓	FID-I3D↓	FVD↓	rank↑	ρ_s ↑	rank [95% CI]↑	ρ_s ↑	Yield
Loss	conditioning only	**17.73**	**0.710**	**0.157**	**159.2**	193.5	0.981	0.913	**0.512** [.38,.64]	**+0.024**	0.956
	+ $\mathcal{L}_{\text{dyn}}$	16.58	0.688	0.185	162.5	**192.7**	0.975	0.912	0.508 [.39,.63]	+0.006	0.959
	+ $\mathcal{L}_{\text{ratio}}$	17.06	0.697	0.173	164.6	200.4	0.979	**0.921**	**0.512** [.37,.65]	-0.003	0.975
	+ both ($\mathcal{L}_{\text{dyn}}, \mathcal{L}_{\text{ratio}}$)†	16.69	0.689	0.178	160.7	194.9	**0.985**	0.909	0.502 [.37,.63]	-0.033	0.959
Cond.	FiLM	16.49	0.680	0.180	171.7	204.6	0.977	0.909	0.492 [.37,.62]	-0.003	0.953
	Raw S_C (no Fourier)	15.82	0.674	0.190	161.5	192.3	0.981	0.911	**0.514** [.37,.66]	-0.013	0.934
Schedule	Joint training	16.58	0.687	0.178	163.9	193.7	0.973	0.911	**0.527** [.40,.65]	**+0.037**	0.959
	Perceptual pretrain	**16.82**	**0.694**	**0.172**	156.0	187.9	**0.981**	**0.918**	0.498 [.37,.63]	+0.012	0.944
	Pretrain + phys. fine tune†	16.75	0.690	0.176	**151.6**	**176.2**	0.975	0.906	0.500 [.37,.62]	-0.022	0.953

attrition is material. Sweeping $r_{t,\min} \in \{0.30, \ldots, 0.90\}$ with an upper bound 1.35 keeps slip rank in 0.42–0.54 (mean 0.51), so rank order is not threshold-driven. Details are in Appendix A3.4. Because the same analytical kinematics define targets and inversion, the evaluation can still reward videos that produce the right centroid orbit without proving true self-spin or contact physics. We address this by keeping training and scoring signals disjoint, reporting gate yield, and treating rank scores as kinematic controllability rather than physical truth.

6.2 Real regime design study

Table 1 varies one factor at a time around the full model: cross attention, both physics losses, K=16, B_{eff}=8, and two-phase training. The full model is rerun per group, so small differences reflect five-seed variance.

Speed is robust in R: every noncollapsed variant reaches rank ≥ 0.97. Slip stays at chance with CIs spanning 0.5, as expected for unmeasured S_C restricted to $[0.92, 0.98]$. The "conditioning only" row removes $\mathcal{L}_{\text{dyn}}, \mathcal{L}_{\text{ratio}}$ but keeps RPM and S_C tokens; its speed rank 0.981 shows that conditioning already steers speed, while physics losses give the best loss-group speed rank (0.985) and the best S FID-I3D/FVD below. Conditioning and schedule mainly tune appearance. Supplementary sweeps show stable speed rank for $B_{\text{eff}} \in \{2, 4, 8, 16\}$ and $K \in \{4, 8, 16\}$, with collapse at K=32.

6.3 Synthetic slip identifiability

Regime S tests slip identifiability with informative labels. Table 2 shows speed rank ≥ 0.99 and slip rank ≥ 0.735 for all settings, with CIs excluding chance.

Cross attention alone exploits the wide synthetic slip grid, so physics losses do not raise slip rank on S; they improve distributional quality for the combined model (FID-I3D 236.4, FVD 116.0). The "conditioning only" row is not image-only: it still receives Fourier RPM and slip tokens, and clean synthetic labels make the ordinal slip axis easy for conditioning alone. The losses optimize noisier

Table 2: Synthetic loss sweep (S), five-seed mean, plausibility gate, known $S_C \in \{0.20, 0.40, 0.60, 0.80\}$. Bold marks best per column. Yield is the fraction of clips passing the physical validity gate.

Losses	PSNR↑	SSIM↑	LPIPS↓	FID-I3D↓	FVD↓	Speed rank↑	Slip rank [95% CI]↑	Slip ρ_s ↑	Yield
conditioning only	21.39	0.733	0.203	243.7	126.9	**0.996**	**0.769** [.63,.90]	**+0.615**	0.678
$+ \mathcal{L}_{\text{dyn}}$	**21.61**	0.719	**0.197**	246.1	127.9	0.992	0.736 [.59,.86]	+0.543	0.669
$+ \mathcal{L}_{\text{ratio}}$	21.36	0.723	0.200	245.5	126.4	**0.996**	0.766 [.63,.88]	+0.550	0.688
$+\mathcal{L}_{\text{dyn}} + \mathcal{L}_{\text{ratio}}$	21.58	**0.734**	0.201	**236.4**	**116.0**	**0.996**	0.735 [.59,.87]	+0.507	0.666

Table 3: Leave-one-speed-out generalization, five-seed mean, plausibility gate. R has unobserved slip; S has known $S_C \in \{0.20, 0.40, 0.60, 0.80\}$.

		Visual				**Speed**	**Slip**	
Held out	**Type**	PSNR↑	SSIM↑	LPIPS↓	FID-I3D↓	rank [95% CI]↑	rank [95% CI]↑	Yield
Regime R (real polishing; GT slip unobservable)								
21 RPM	boundary	11.66	0.540	0.315	**147.1**	0.631 [.52,.74]	0.492 [.34,.64]	0.884
48 RPM	interior	11.81	0.547	0.307	144.4	0.827 [.75,.90]	0.487 [.34,.64]	0.822
78 RPM	interior	**12.78**	**0.581**	**0.279**	175.0	**0.875** [.81,.93]	**0.588** [.44,.73]	0.922
108 RPM	boundary	12.01	0.561	0.298	169.7	0.640 [.61,.67]	0.485 [.36,.60]	**0.981**
Regime S (synthetic; GT slip known)								
21 RPM	boundary	20.44	**0.725**	0.238	**228.2**	0.696 [.42,.92]	0.680 [.44,.89]	0.416
48 RPM	interior	20.12	0.670	0.241	253.9	**0.938** [.86,.99]	0.722 [.56,.86]	0.634
78 RPM	interior	19.75	0.660	0.250	249.5	0.721 [.62,.82]	**0.742** [.62,.86]	0.709
108 RPM	boundary	**20.48**	0.719	**0.235**	233.0	0.665 [.65,.68]	0.723 [.60,.83]	**0.884**

surrogates: $\mathcal{L}_{\text{dyn}}$ is speed-dominated, while $\mathcal{L}_{\text{ratio}}$ depends on D^{lat}, which is more sensitive to total visible displacement than to slip-specific circulation or self-spin. With finite batches, this can trade a few rank points for better video-distribution metrics. Learned directional flow or video features may help, but scalar flow magnitude is insufficient because it measures the amount of motion, not whether motion follows $R_T(S_C)$ (Appendix A3.2). Thus, slip is learnable when exposed; the real bottleneck is S_C observability.

6.4 Leave one speed out

We hold out one RPM during training and evaluate the full model on the held-out fold. Interior folds (48, 78 RPM) test interpolation; boundary folds (21, 108 RPM) test extrapolation.

Both regimes show easier speed interpolation than boundary extrapolation. Slip diverges by regime: real LOSO remains at chance, while synthetic LOSO is above chance on all folds and excludes chance on three of four.

6.5 Non-generative flow baselines

Two training-free flow-warping baselines test whether simple kinematics can replace generation. **GT-match** warps the first real frame with sinusoidal displacement matched to ground-truth flow magnitude at each RPM, making slip flat by construction. **Analytical** uses $\omega_c = \omega_p(1+S_C \sin\theta)/(1+\sin\theta)$ from Eq. 1, activating slip tautologically. The baselines are deterministic sanity checks on $S_C \in \{0.30, 0.50, 0.70, 0.95\}$. To avoid mixing protocols, our model row reuses the five-seed real-regime full-model result from Table 1; real slip therefore remains at chance in both tables.

Our model has higher paired fidelity than flow baselines (PSNR 16.69 vs. 13.60/12.37), but real slip rank is chance (0.502, CI spanning 0.5), matching Table 1. The analytical ceiling (0.804) is tautological and inaccessible without prescribing the true orbit. Flow baselines win FID-I3D/FVD because they warp real frames; PSNR/SSIM/LPIPS favor the generator. The real-data takeaway is speed controllability and visual fidelity, not demonstrated slip controllability.

7 Discussion

The experiments separate a positive control from an identifiability limit. Speed control is reliable: noncollapsed R variants reach rank ≥ 0.97, and the same recipe reaches ≥ 0.99 in S. A useful

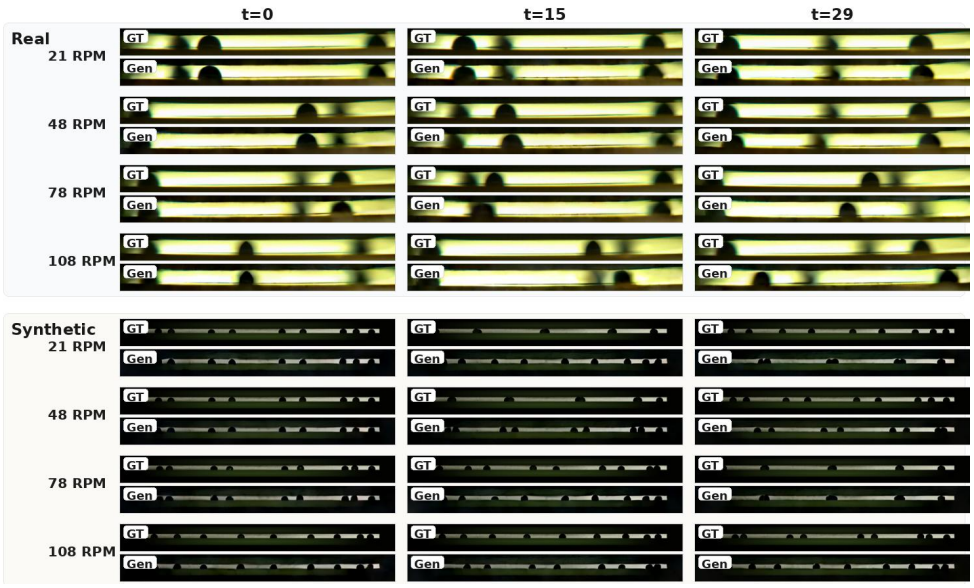


Figure 5: Representative RPM sweep for the full model. Displacement increases with RPM, consistent with the robust speed axis ranking in Table 1; slip effects are subtler and are evaluated through tracking and Lee inversion. Appendix A3.6 shows every frame of a generated clip at each speed for both regimes.

Table 4: Non-generative flow baselines and the five-seed real-regime model result. Baselines are deterministic wide-slip sanity checks; the model row is the same narrow-prior real result as Table 1, including its gate yield of 0.959.

Method	PSNR↑	SSIM↑	LPIPS↓	FID-I3D↓	FVD↓	Slip rank [95% CI]↑
GT-match (flow)	13.60	0.679	0.248	**41.8**	37.9	0.484 (at chance)
Analytical (kinematic)	12.37	0.591	0.306	43.3	**26.1**	0.804 (tautological†)
Ours (full model, 5 seeds)	**16.69**	**0.689**	**0.178**	160.7	194.9	0.502 [.37,.63]

† The analytical baseline drives the warp with the same kinematic equation the scorer inverts, making its rank accuracy a tautological ceiling, not an achievable target.

envelope is $K \in \{4, 8, 16\}$, cross attention, and effective batch sizes 4–16; the schedule mainly tunes FID-I3D/FVD. Slip control depends on labels: S reaches rank ≥ 0.735 with CIs excluding chance, while R stays at chance because S_C is unmeasured and restricted to $[0.92, 0.98]$. We claim slip learnability under informative labels, not the transfer of synthetic slip control to real polishing.

Main limitations are a small number of video clips 604 from one real regime, fewer passing plausibility cells in R than in S, weaker boundary LOSO generalization, and lower held-out-speed appearance. The surrogate also stops short of being a full process-optimization tool: it predicts plausible kinematic videos, not material removal, roughness, or defect formation. Its immediate use is exploratory design-of-experiments support, sanity-checking operating sweeps, and stress-testing downstream trackers; closed-loop polishing decisions would require independent metrology linking generated kinematics to surface outcomes. Boundary LOSO failures are consistent with extrapolating both appearance and speed statistics beyond the observed range; interior folds mostly test interpolation and are more stable. This framing keeps the practical claim modest under current sensing constraints and data: the model is useful for asking whether a requested operating change produces plausible video kinematics, not for certifying a polishing recipe by itself.

Self-spin ω_b and unobservable DOF. The kinematic model predicts orbit (ω_c) and self-spin (ω_b). Our centroid tracker recovers ω_c, but ω_b appears mainly as surface texture and needs multi-angle cameras or model-based texture analysis. Since D^{lat} tracks pixel motion (r=0.91) but emphasizes circulation, the model is a kinematic video surrogate, not a contact-mechanics or material-removal simulator.

$K{=}32$ **collapse.** With only four training RPMs, $K{=}32$ introduces harmonics at $k \geq 16$ that exceed the four-point Nyquist rate, encouraging grid overfit and collapsed speed rank. $K{=}16$ is the recommended default.

Compute and scalability. Each run costs 12–16 GPU-hours on one RTX 4090; inference takes 35–45 s per 30-frame $496{\times}48$ clip. Rank-32 LoRA adds $\approx$ 22M trainable parameters to frozen 1.5B-parameter SVD-xt. Larger slip grids or geometries need more data, but only the kinematics must be re-parameterized.

Future directions. An EM loop alternating latent slip estimation with model updates may reduce the need for per-clip slip measurement; S shows the generator can recover slip when labels exist. Learned keypoints could recover both ω_c and ω_b, and the renderer enables virtual sweeps over groove angle, capsule size, and speed before physical trials.

Broader impacts. The work targets manufacturing video surrogates for ICF capsule polishing, where controllable generation may reduce process tuning costs. It does not model implosion physics or add information beyond public polishing kinematics, but ICF has dual-use exposure and release plans that should be reviewed accordingly. The main misuse risk is treating generated clips as physical evidence beyond the scope; we separate perceptual and kinematic metrics, state that real slip is unmeasured, and require independent validation before closed-loop use.

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Appendix

A1 Kinematic Equations

This supplementary section derives Eqs. (1)–(2) from first principles, following Lee et al. [26, 27]. The analysis assumes a rigid ball of radius R_b resting in a concentric V-groove of half-angle θ and groove radius R_g cut into a baseplate spinning at ω_p, with an upper lap loaded against the ball. Contact A is the inner groove wall, B the outer groove wall, and C is the head contact with the upper lap.

A1.1 Geometry

The ball contacts A and B at radial distances

$$R_i \;=\; R_g - R_b\cos\theta, \qquad R_o \;=\; R_g + R_b\cos\theta, \tag{8}$$

from the baseplate axis, and C at a radial distance R_g. The ball's spin axis tilts by β with $\tan\beta = R_b\sin\theta/R_g$; β depends only on geometry.

A1.2 No-slip at the groove walls

Let ω_c be the circulation angular speed about the baseplate axis and ω_b the self-spin. Linear contact velocities at A and B from baseplate motion must be balanced by ball-surface velocities:

$$R_i\omega_p = R_i\omega_c + R_b\omega_b\sin(\theta-\beta), \qquad R_o\omega_p = R_o\omega_c + R_b\omega_b\sin(\theta+\beta). \tag{9}$$

These are the no-slip conditions at A and B.

A1.3 Slip at the head contact

At contact C,

$$V_C \;=\; R_g\,\omega_{c,\mathrm{head}} - R_g\,\omega_c \;=\; S_C\,R_g\,\omega_p, \tag{10}$$

where $\omega_{c,\mathrm{head}}$ is the no-slip ball-circulation speed and $S_C\in[0,1]$ is the dimensionless head-contact slip ratio. Combining Eq. (8)–(10) and using standard trigonometric identities yields

$$\omega_c = \omega_p\,\frac{1+S_C\sin\theta}{1+\sin\theta}, \qquad \omega_b = \frac{R_g\,\omega_p\,(1-S_C)}{R_b\cos\beta\,(1+\sin\theta)}, \tag{11}$$

which, together with Eq. (10), gives the three components of Eq. (1).

A1.4 Effective sliding speed and slip-ratio target

Aggregating the three component *surface speeds*, $V_{\text{eff}} \triangleq \sqrt{(R_g\omega_c)^2 + (R_b\omega_b)^2 + V_C^2}$ is the first half of Eq. (2). Multiplying the angular velocities by their physical radii keeps all terms in m/s, so the target is invariant to the choice of length unit. Since each component speed scales linearly with ω_p for fixed S_C, V_{eff} is linear in ω_p with zero intercept and is strictly increasing in ω_p, which is the linearity used by $\mathcal{L}_{\mathrm{dyn}}$ (Eq. 4). The circulation-to-plate ratio $R_T(S_C) = \omega_c/\omega_p = (1+S_C\sin\theta)/(1+\sin\theta)$ is strictly increasing in S_C for $\theta\in(0,\pi/2)$, providing the analytical ordinal target used by $\mathcal{L}_{\mathrm{ratio}}$ (Eq. 5).

A2 Theory Derivations

This supplementary section backs Sec. 5: (i) well-posedness and asymptotic calibration of $\mathcal{L}_{\mathrm{dyn}}$ (Sec. 5.1); (ii) the variance-decomposition argument for why $\mathcal{L}_{\mathrm{ratio}}$ is stratified (Sec. 5.2); and (iii) the latent–pixel consistency diagnostic, with the empirical instantiation $r{=}0.91$ on $n{=}120$ clips (Sec. 5.3).

A2.1 Well-posedness and asymptotic calibration of $\mathcal{L}_{\mathrm{dyn}}$

Setup. Write $V_i \triangleq V_{\text{eff}}(\omega_p^{(i)}, S_C^{(i)})$ and $D_i \triangleq D_i^{\mathrm{lat}}$ (Eq. 4). The finite-batch non-degeneracy hypotheses are $s_{B,V}^2>0$ and $s_{B,D}^2>0$, where $s_{B,V}^2 \triangleq (B-1)^{-1}\sum_i(V_i-\bar V)^2$ and analogously for D. A sufficient practical condition for $s_{B,V}^2>0$ is that the batch contain at least two realized speeds V_{eff}; our stratified sampler enforces this by placing multiple physical RPM strata in each batch.

Well-posedness. The batch-Pearson correlation is

$$\hat r_B \;=\; \frac{\sum_{i=1}^{B}(V_i-\bar V)(D_i-\bar D)}{\sqrt{\sum_i(V_i-\bar V)^2}\,\sqrt{\sum_i(D_i-\bar D)^2}}. \tag{12}$$

The empirical variances $\frac{1}{B}\sum_i(V_i-\bar V)^2$ and $\frac{1}{B}\sum_i(D_i-\bar D)^2$ are strictly positive for $B\ge 2$ under the hypotheses above, so the denominator is positive and $\hat r_B\in[-1,1]$ is well-defined. $\mathcal{L}_{\mathrm{dyn}} = -\hat r_B$

inherits well-posedness. The numerator and denominator are locally Lipschitz functions of $\{D_i\}$ away from the zero-variance set. If the denoiser outputs are locally Lipschitz functions of the parameters, the composition is locally Lipschitz in the denoiser parameters away from the zero-variance set. Because each D_i is an ℓ_1 latent frame-difference, differentiability is almost everywhere: the only additional excluded set is where an adjacent latent pixel difference is exactly zero.

Asymptotic calibration via Fisher's z-transform. For bivariate-normal (V, D) with correlation $\rho \in (-1, 1)$, Fisher's z-transform $z_F(r) = \frac{1}{2} \log \frac{1+r}{1-r} = \tanh^{-1}(r)$ satisfies, classically [see e.g. 11],

$$\sqrt{B-3}\left(z_F(\hat{r}_B) - z_F(\rho)\right) \xrightarrow{d} \mathcal{N}(0,1) \quad \text{as } B \to \infty. \tag{13}$$

An asymptotic $(1-\delta)$-CI on the z_F-scale therefore has a half-width $z_{1-\delta/2}/\sqrt{B-3}$.

Setting this asymptotic half-width to be at most ε corresponds to the reference size $B^* = \left\lceil 3 + z^2_{1-\delta/2}/\varepsilon^2 \right\rceil$. For ε=0.1, δ=0.05, $z_{1-\delta/2}$=1.96, so B^*=$\lceil 3 + (1.96)^2/0.01 \rceil = 388$.

With only finite fourth moments, the sample covariance vector is still asymptotically normal by the multivariate CLT and $\hat{r}_B$ follows by the delta method [41], but the variance is distribution-dependent. Thus the $1/(B-3)$ formula is a Gaussian reference calculation, not an exact finite-batch interval for our four-RPM grid.

Stratified-batching corollary. If $\{\omega_p^{(i)}\}_{i=1}^B$ is constructed so each batch contains at least one sample in each of a fixed set of ω_p-strata, the empirical variance $\frac{1}{B}\sum_i (V_i - \bar{V})^2$ is bounded below by the between-strata variance of V_{eff}, a deterministic constant $\sigma^2_{\min} > 0$. This tightens the denominator of $\hat{r}_B$ in the finite batch and prevents collapse of the correlation objective. We use this corollary only as a non-degeneracy argument for the training loss, not as a claim that the resulting batch correlation is an unbiased or lower-variance estimator of a population correlation. Our experimental practice of stratified sampling across the four speeds is consistent with this corollary. □

A2.2 Variance decomposition for the stratified $\mathcal{L}_{\text{ratio}}$

Setup. A training batch $\mathcal{B}$ of size B partitions into RPM strata $\{\mathcal{B}_r\}_{r \in \mathcal{R}_B}$ with $|\mathcal{B}_r| \geq 2$. Within each stratum, samples share ω_p=r and have i.i.d. $S_C^{(i)} \sim \text{Beta}(\alpha, \beta)$ with μ_{S_C}=$\alpha/(\alpha+\beta)$ and $\sigma^2_{S_C} = \alpha\beta/[(\alpha+\beta)^2(\alpha+\beta+1)]$. For $\text{Beta}(199.5, 10.5)$, μ_{S_C}=0.95, σ_{S_C}=0.0150, nominal high-density range approximately $[0.92, 0.98]$.

Variance of $R_T(S_C)$ within and across strata. $R_T(S_C)$ is affine in S_C with slope $m \triangleq \sin\theta/(1+\sin\theta) \approx 0.414$ at θ=45°, so

$$\text{Var}(R_T)_{\text{within}} \;=\; m^2 \sigma^2_{S_C} \;\approx\; 3.86 \times 10^{-5}. \tag{14}$$

Because R_T depends on S_C but not directly on ω_p, pooling RPM strata does not substantially increase the variance of R_T when the same S_C prior is used in each stratum. The problem is instead that D^{lat} inherits a much larger speed-driven component: V_{eff} varies by ∼5× from 21 to 108 RPM, whereas the within-stratum variation of R_T under the real Beta prior is only $m^2\sigma^2_{S_C} \approx 3.86 \times 10^{-5}$. Decomposing $D^{\text{lat}} = \alpha_v V_{\text{eff}} + \alpha_R R_T + \xi$ with mean-zero ξ,

$$\text{Cov}_{\text{pooled}}(D^{\text{lat}}, R_T) = \alpha_R \,\text{Var}(R_T)_{\text{pooled}} + \alpha_v \,\text{Cov}(V_{\text{eff}}, R_T)_{\text{pooled}}. \tag{15}$$

When S_C is sampled independently of RPM, $\text{Cov}(V_{\text{eff}}, R_T)_{\text{pooled}}$ is small in expectation relative to the RPM-driven variance in V_{eff}, while the Pearson denominator for D^{lat} is dominated by $\alpha_v^2 \,\text{Var}(V_{\text{eff}})_{\text{pooled}}$. Thus the pooled correlation can underweight the weak slip-axis variation, motivating the within-RPM stratified loss.

Mean-normalized within-RPM stratification. Let $\bar{D^{\text{lat}}}_{\mathcal{B}_r} \triangleq |\mathcal{B}_r|^{-1} \sum_{i \in \mathcal{B}_r} D_i^{\text{lat}}$ and $\bar{R_T}_{\mathcal{B}_r}$ analogously, and define $\tilde{D}^{(i)} = D_i^{\text{lat}} / \bar{D^{\text{lat}}}_{\mathcal{B}_r}$, $\tilde{R}^{(i)} = R_T(S_C^{(i)}) / \bar{R_T}_{\mathcal{B}_r}$. Within a stratum V_{eff} is nearly constant because the RPM is fixed and the remaining S_C-dependence through V_C, ω_c, and ω_b is small under σ_{S_C}=0.015. So, the dominant axis of within-stratum variation in D^{lat} is whatever S_C-dependence the model produces. Mean-normalization removes the absolute scale, and $\mathcal{L}_{\text{ratio}}$ (Eq. 5) penalizes within-stratum order mismatches between latent dynamics and analytical R_T.

- *Why MSE rather than Pearson within the stratum?* For $|\mathcal{B}_r|=2$, sample Pearson is ± 1 with probability 1 and its gradient is degenerate; MSE between mean-normalized signals stays smooth.
- *Why mean-normalization rather than centering?* Both remove the across-stratum offset; mean-normalization additionally normalizes within-stratum scale, keeping $\mathcal{L}_{\text{ratio}}$ dimensionless and comparable across strata. Centering would re-introduce an RPM-dependent weight per stratum.

Filtering criterion. We exclude strata with $|\mathcal{B}_r| < 2$ or $\text{Var}_{\mathcal{B}_r}(R_T) \leq \varepsilon_{S_C}$, since these cannot contribute slip information. The stratified form is therefore a design constraint that prevents RPM-dominated variance from swamping the slip gradient at our batch size, not a separate proof of slip recovery.

A2.3 Latent–pixel consistency diagnostic

Setup. Let $\mathcal{E} : \mathbb{R}^{T\times 3\times H\times W} \to \mathbb{R}^{T\times C\times h\times w}$ be the frozen VAE encoder. We define

$$
\begin{aligned}
D_i^{\text{pix}} &= \frac{1}{(T-1)\cdot 3HW} \sum_{t=1}^{T-1} \left\| \mathbf{v}_i^{(t+1)} - \mathbf{v}_i^{(t)} \right\|_1, \\
D_i^{\text{lat}} &= \frac{1}{(T-1)Chw} \sum_{t=1}^{T-1} \left\| \mathbf{z}_i^{(t+1)} - \mathbf{z}_i^{(t)} \right\|_1,
\end{aligned}
\tag{16}
$$

where $\mathbf{z}_i = \mathcal{E}(\mathbf{v}_i)$. Because the diffusion model is trained in latent space, the physics losses use D^{lat} rather than optical flow or pixel differences. The required contract is therefore not pixel-level physical fidelity, but preservation of the ordering of clip-level motion magnitudes.

Diagnostic criterion. We test this contract empirically by measuring the Pearson correlation between D^{pix} and D^{lat} on held-out real clips. A high positive correlation indicates that clips with larger pixel-level temporal change also tend to have larger latent temporal change, making D^{lat} a suitable scalar proxy for the ordering objective used by $\mathcal{L}_{\text{dyn}}$. This diagnostic does not prove that latent dynamics recover contact mechanics; it only verifies that the latent scalar preserves the dominant ordering of visible motion on the real clips used in this study.

Table 5: Empirical Pearson r between D^{pix} and D^{lat} (Sec. 5.3).

Split / band	r	p-value
Overall (n=120, four RPMs pooled)	**0.907**	3.30×10^{-46}
RPM 21 (within-group)	0.778	n/a
RPM 48 (within-group)	0.827	n/a
RPM 78 (within-group)	0.939	n/a
RPM 108 (within-group)	0.920	n/a
Low band (2×2 pooled, circulation signal)	0.985	n/a
High band (residual, self-rotation signal)	0.730	n/a

Empirical instantiation. We sampled n=120 real polishing clips total (30 per RPM) of T=14 frames, encoded them through the frozen SVD-xt VAE, and measured D^{pix} and D^{lat} per clip. Table 5 reports the resulting correlations. As a reproducibility check, recomputing V_{eff} with the surface-speed norm in Eq. (2) preserves the ordering of prior diagnostic runs (Pearson r=0.997 with the earlier audit target), while providing the dimensionally correct physical scale used in this paper.

The overall correlation r=0.907 supports using D^{lat} as a clip-level motion-ordering proxy for $\mathcal{L}_{\text{dyn}}$. The lower high-band correlation r=0.730 also exposes a limitation: latent frame differences preserve circulation like motion more reliably than residual self-rotation. This is why Sec. 5.3 states the result as an empirical scope condition rather than as a guarantee of full contact-mechanics recovery.

A3 Additional Experimental Results

This supplementary section expands the compact results in Sec. 6. The main paper reports the primary design, synthetic, and LOSO results; here we include the effective-batch-size and Fourier-bandwidth sweeps referenced in Sec. 6.2, along with metric definitions and controlled sweep details.

A3.1 Metric definitions

I3D FVD. We use RGB-I3D features trained on Kinetics-400 [40]. Pre-processing is fixed for all variants: resize to 224×224, sample 16 frames per clip by bilinear time interpolation from the 30-frame videos, and compute the Fréchet distance between generated and reference feature distributions.

FID and paired image metrics. FID [18] is computed with the same RGB-I3D backbone used for FVD so relative comparisons do not mix feature spaces. PSNR, SSIM, and LPIPS [50] are paired against the held-out clip associated with the same first frame.

Diagnostic flow correlation. A diagnostic Pearson correlation between RPM and mean per-clip Farneback flow magnitude [10] is computed for sanity-checking, but the headline speed and slip metrics are the gated rank accuracies recovered through the tracking pipeline (Sec. 6.1).

A3.2 Alternative motion signals for $\mathcal{L}_{\mathrm{dyn}}/\mathcal{L}_{\mathrm{ratio}}$

We also checked optical-flow magnitude as a candidate dynamics signal. Table 6 summarizes the diagnostic correlations and controlled slip-ranking baselines. On the same 120 real clips used for the latent–pixel diagnostic, Farneback mean flow magnitude correlates strongly with pixel L1 (r=0.933), latent L1 (r=0.833), and the corrected surface-speed V_{eff} (r=0.905). Thus flow is a good aggregate-motion diagnostic. It does not by itself solve slip supervision, however: a scalar flow magnitude measures how much the image moves, not whether the motion follows the slip-sensitive circulation ratio $R_T(S_C)$.

Table 6: Alternative motion-signal diagnostics for $\mathcal{L}_{\mathrm{dyn}}$ and $\mathcal{L}_{\mathrm{ratio}}$.

Signal / baseline	**Diagnostic**	**Value**
Farneback flow vs. D^{pix}	Pearson r on real clips	0.933
Farneback flow vs. D^{lat}	Pearson r on real clips	0.833
Farneback flow vs. corrected V_{eff}	Pearson r on real clips	0.905
Flow-magnitude warp	Controlled slip rank accuracy	0.484
Analytical kinematic warp	Controlled slip rank accuracy	0.804

The flow-magnitude warp is intentionally slip-flat: it matches the mean flow magnitude at each RPM and therefore answers the reviewer concern that the model might only be learning speed. Its rank accuracy is at chance. The analytical warp uses the Lee kinematics directly and activates the slip axis, but it is a hand-coded image warp and has weaker paired fidelity than the five-seed diffusion model row in Table 4 (PSNR 12.37 and SSIM 0.591 versus 16.69 and 0.689). Learned flow or video embeddings such as RAFT [37] and VideoMAE [39] are plausible future signals if they retain direction and track geometry, but a scalar learned flow magnitude would likely inherit the same limitation as Farneback magnitude. Complete $\mathcal{L}_{\mathrm{dyn}}/\mathcal{L}_{\mathrm{ratio}}$ variants with these proxies would require decoding latents or adding a frozen video encoder inside the training loop and then calibrating a vector feature to scalar physics targets. The latent ℓ_1 signal keeps the loss cheap, differentiable almost everywhere, and aligned with the SVD latent space while the analytical V_{eff}, R_T targets provide the missing physical direction.

A3.3 Tracking and kinematic inversion pipeline

The evaluation pipeline translates generated video clips into estimated kinematic quantities $(\widehat{\omega_c}, \widehat{S_C})$ using only pixel observations, with no access to latent codes or training signals. This independence is essential: a pipeline that reused training-time quantities would not constitute an external physical test of controllability.

Step 1: Detection and tracking. A YOLO11 detector fine-tuned on real polishing videos localises shell centroids in each frame. Detections are linked across frames by a Mahalanobis-IOU hybrid tracker following [52], which uses detection confidence to weight the association cost. Each track k is a sequence of frame-indexed centroids and uncertainty estimates $\{(t_n,\, c_{x,n},\, \sigma_{x,n})\}_{n=1}^{N_k}$.

Step 2: Per-track angular velocity fitting. Because the shell orbits in a plane, its x-centroid follows a cosine projection of the orbit angle. Inverting that projection and fitting the unwrapped angle

linearly in time gives an unbiased angular speed estimate without requiring the full 2-D orbit to be visible.

For track k, project the centroid onto the orbit angle:

$$u_n = \frac{c_{x,n} - x_{\mathrm{ctr}}}{r_{\mathrm{orb}}}, \qquad \theta_n = \cos^{-1}(u_n), \tag{17}$$

where x_{ctr} and r_{orb} are the orbit centre and radius estimated from the suite geometry. The raw $\arccos$ introduces a reflection ambiguity at the orbit poles; this is resolved by a sign-of-Δx unwrapping step that gives a monotonically increasing angle series $\tilde{\theta}_n$. A weighted ordinary-least-squares fit then yields the signed angular speed:

$$\hat{\omega}^{(k)} = \arg\min_{a,b} \sum_n \big(\tilde{\theta}_n - a t_n - b\big)^2. \tag{18}$$

The fit quality is assessed by $R^2 = 1 - \mathrm{SS}_{\mathrm{res}}/\mathrm{SS}_{\mathrm{tot}}$. Four per-track gates remove unreliable estimates before aggregation: (i) minimum track length $N_k \geq N_{\min}$; (ii) near-singularity fraction $|\{n : |u_n| > u^*\}|/N_k \leq f^*$ (tracks that spend most of their time near the arccos poles have large projection noise); (iii) minimum angular range $\max_n \tilde{\theta}_n - \min_n \tilde{\theta}_n \geq \Delta\theta_{\min}$; (iv) minimum fit quality $R^2 \geq R^2_{\min}$.

Step 3: Cross-track aggregation. Even after per-track gating, a small number of spurious tracks can bias a simple mean. We therefore apply a two-stage robust aggregation [34].

First, a median absolute deviation (MAD) outlier screen removes tracks whose slope deviates from the per-clip median by more than c scaled MADs:

$$\mathrm{MAD} = \mathrm{median}_k\big(|\hat{\omega}^{(k)} - \mathrm{median}_k\, \hat{\omega}^{(k)}|\big)\,, \ \text{keep track } k \iff \big|\hat{\omega}^{(k)} - \mathrm{median}_k\, \hat{\omega}^{(k)}\big| \leq c \cdot 1.4826 \cdot \mathrm{MAD} \tag{19}$$

Second, a weighted-median aggregation over the surviving set $\mathcal{K}$ uses weights $w_k = N_k \cdot \max(R_k^2, 0)$ to up-weight long, well-fit tracks:

$$\widehat{\omega_c} = \mathrm{sgn}(\text{majority}) \cdot |\hat{\omega}_{(\pi)}|, \quad \pi = \min\Big\{k : \sum_{j \leq k} w_{(j)} \geq \tfrac{1}{2} \textstyle\sum_j w_j\Big\}, \tag{20}$$

where (j) indexes tracks sorted by $|\hat{\omega}|$ and the majority sign is determined by sign-consistency voting across $\mathcal{K}$.

Step 4: Physical validity gate. The kinematic inversion in Step 5 is only interpretable when the recovered $\widehat{\omega_c}$ lies within the physically plausible range implied by the Lee model [26] for the suite's operating conditions. Clips whose $\widehat{\omega_c}$ falls outside this window are excluded from rank accuracy; reporting only on plausible clips prevents out-of-regime tracker noise from polluting the controllability scores.

From the Lee kinematics, the circulation ratio is

$$R_T(S_C, \theta) = \frac{1 + S_C \sin\theta}{1 + \sin\theta}, \qquad \omega_c = \omega_p \cdot R_T, \tag{21}$$

giving a plausible window

$$0.5\, \omega_{p,\min} R_{T,\min} \ \leq \ |\widehat{\omega_c}| \ \leq \ \kappa\, \omega_{p,\max} R_{T,\max}, \tag{22}$$

where $R_{T,\min\,/\,\max}$ are evaluated at the suite's slip and RPM extremes. A clip passes the gate when Eq. (22) holds. Averaged over five seeds, clip-level gate yield ranges from 93.4% to 97.5% across the nine real-regime variants in Table 1 and from 66.6% to 68.8% across the four synthetic-regime variants in Table 2, which now report yield per row. This is distinct from the within-clip track-retention rate above (the fraction of individual tracks, not clips, kept after the length, singularity, R^2, and cross-track MAD checks), which is lower, around 28%–32% in the real regime; an earlier draft reported this track-retention figure as gate yield, which was an error.

Step 5: Kinematic inversion. Given the aggregated $\widehat{\omega_c}$ and the known operating speed ω_p, the Lee equations are inverted to recover the circulation ratio and then the contact slip:

$$\widehat{R_T} = \frac{\widehat{\omega_c}}{\omega_p}, \qquad \widehat{S_C} = \frac{\widehat{R_T}\,(1 + \sin\theta) - 1}{\sin\theta}. \tag{23}$$

A single scalar calibration coefficient c_{suite}, estimated on a validation split without test labels, corrects small geometric bias in $\widehat{\omega_c}$ before inversion; no slip labels enter this correction.

Step 6: Rank accuracy. A surrogate is controllable along a given axis if it can *rank* operating conditions correctly, even if absolute calibration is imperfect. Rank accuracy is therefore the natural metric: it equals 1 when generated clips are perfectly ordered by the conditioning variable, 0.5 at chance, and is insensitive to monotone rescaling of $\hat{y}$.

Formally, for a group g of clips sharing the confounding variables (i.e., same scene and slip for the speed axis; same scene and RPM for the slip axis), we compute the pairwise concordance [25]:

$$\mathrm{RankAcc}_g = \frac{\left|\{(i,j) : i<j,\ (\mathrm{gt}_i - \mathrm{gt}_j)(\hat{y}_i - \hat{y}_j) > 0\}\right|}{\left|\{(i,j) : i<j,\ \mathrm{gt}_i \neq \mathrm{gt}_j\}\right|}. \tag{24}$$

The two evaluation axes use different ground-truth and estimate pairs:

Axis	Group by	gt	$\hat{y}$
Speed	(scene, $S_{C,\mathrm{gt}}$)	ω_p (RPM)	$\widehat{\omega_c}$
Slip	(scene, RPM)	$R_{T,\mathrm{gt}}$	$\widehat{R_T}$

The reported value is the bootstrap mean of $\{\mathrm{RankAcc}_g\}$ over groups, with 95% percentile-bootstrap confidence intervals (1000 resamples).

A3.4 Controlled slip sweep and physical-validity gate

For controllability, each variant generates 64 clips per seed: four scenes, four RPM values, and four slip values ($S_C \in \{0.92, 0.94, 0.96, 0.98\}$ on R, $\{0.20, 0.40, 0.60, 0.80\}$ on S). The full tracking and inversion procedure of Sec. A3.3 is run identically on every generated clip. A track is physically valid (plausibility gate, Eq. (22)) when $\widehat{R_T} = \widehat{\omega_c}/\omega_p \in [0.8, 1.2]$. The gate is intentionally conservative: it reduces sample size but prevents tracks outside the Lee-model regime from driving the controllability scores.

A3.5 Effective-batch-size and Fourier-bandwidth sweeps

Table 7 reports the two recipe sweeps referenced in Sec. 6.2.

Table 7: Effective-batch-size and Fourier-bandwidth sweeps on R, mean over five seeds, plausibility gate. **Bold** marks the best entry within each group. Default configuration is marked †.

		Perceptual fidelity					Speed axis		Slip axis	
Group	Variant	PSNR↑	SSIM↑	LPIPS↓	FID↓	FVD↓	rank↑	ρ_s ↑	rank [95% CI]↑	ρ_s ↑
B_{eff}	2	17.52	0.714	0.166	162.0	192.7	0.975	0.907	0.492 [.36,.62]	-0.030
	4	**17.68**	**0.715**	**0.166**	161.0	**188.9**	0.977	0.915	**0.513** [.38,.64]	**+0.047**
	8†	17.10	0.700	0.173	158.0	192.6	0.967	0.908	0.492 [.36,.63]	+0.004
	16	16.38	0.684	0.181	**156.9**	192.4	0.975	**0.916**	0.479 [.35,.61]	-0.010
K	4	16.54	0.685	0.180	160.2	190.3	0.969	0.902	0.498 [.38,.62]	**+0.032**
	8	15.95	0.671	0.193	159.1	**186.2**	0.971	0.900	0.503 [.38,.63]	+0.026
	16†	**16.70**	**0.690**	**0.174**	161.9	194.4	0.971	**0.918**	**0.517** [.38,.65]	+0.021
	32	10.88	0.439	0.356	219.0	264.5	0.770	0.624	0.441 [.24,.65]	-0.098

A3.6 Qualitative generated frames across speeds

Sec. 6 reports frame-level and video-level metrics but shows only one representative frame per condition (Fig. 5). To let a reader judge temporal coherence directly, Figs. 6 and 7 show every frame of a generated clip at each of the four RPMs, for the full model (both physics losses, cross attention, pretrain plus physics fine tune) at its final training epoch. These clips come from the model's saved qualitative sample and validation outputs at the final epoch, at a thin top-down crop resolution (496×48 pixels for R, 480×48 for S) rather than full evaluation resolution. Frames keep their native aspect ratio; each speed block lays its 30 frames out in a 10×3 grid rather than one long row, so individual frames stay large and readable without any stretching. The synthetic videos use

S_C=0.50, a value from training-time monitoring rather than the $\{0.20, 0.40, 0.60, 0.80\}$ evaluation grid in Sec. 6.3. Across both figures, the capsule contact point moves smoothly from frame to frame with no visible flicker or identity swap, and it advances further per frame at higher RPM, consistent with the speed axis ranking in Table 1.

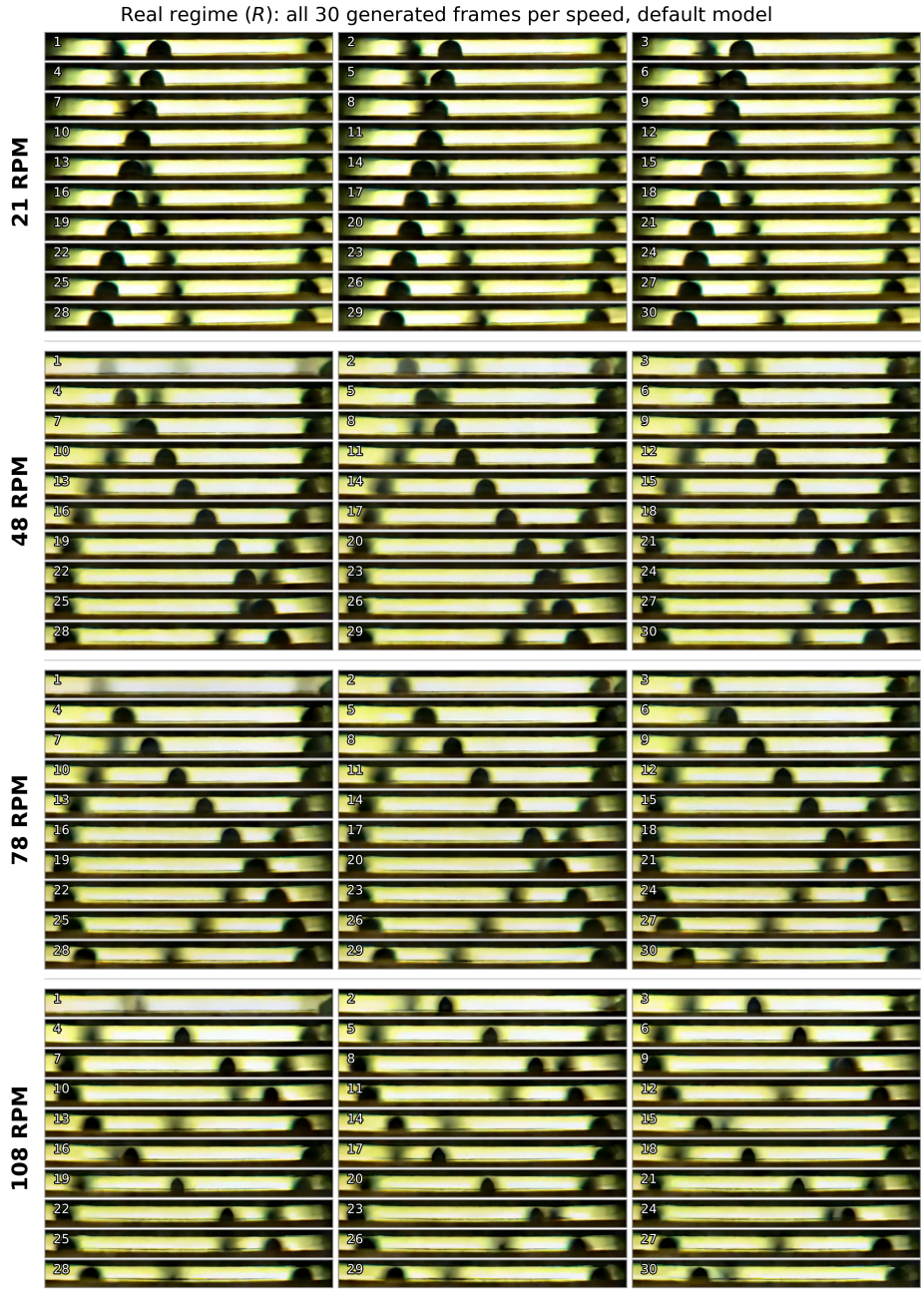


Figure 6: All 30 frames of the full model's generated clip at each real-regime (R) speed, read left to right then top to bottom within each speed block.

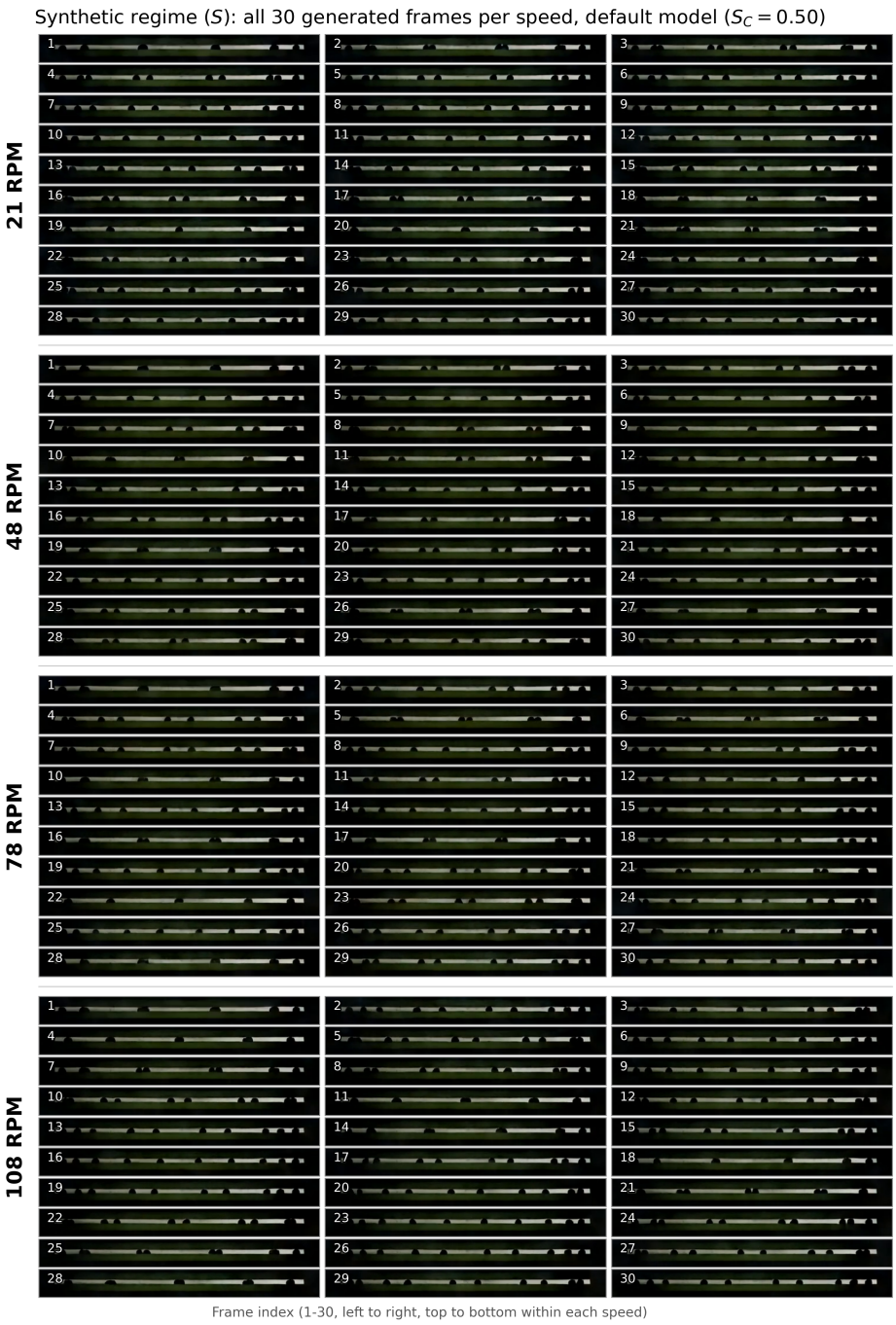


Figure 7: All 30 frames of the full model's generated video at each synthetic-regime (S) speed, at S_C=0.50, read left to right then top to bottom within each speed block.

A4 Training Details

A4.1 Hyperparameters

Table 8 summarizes the training and inference hyperparameters used for the main experiments.

Table 8: Training and inference hyperparameters.

Parameter	Value
Optimization	
Optimizer	AdamW
Weight decay	0.01
LoRA learning rate	1×10^{-4}
Physics-encoder learning rate	5×10^{-4}
LR schedule	Cosine + 500-step warmup
Per-GPU batch size	2 (default real-design runs)
Gradient accumulation	4 steps (default real-design runs)
Effective batch size	8 default; audited per suite below
LoRA rank r	32
Epochs	50 (per ablation script)
GPUs	single RTX 4090 per run; multiple runs in parallel across GPUs
Conditioning	
Number of physics tokens	3 (RPM, S_C, interaction)
Fourier bands K for each scalar	16
CFG drop probability p_{cfg}	0.1
Token dim d	1024 (matches SVD-xt CLIP embedding)
Two-phase schedule	
Phase-1 epochs (perceptual only)	10
Ramp window (linear $w(e)$)	5 epochs
Loss weights (full config)	
λ_r (reconstruction)	1.0
λ_s (SSIM)	0.5
λ_d (dynamics, $\mathcal{L}_{\text{dyn}}$)	0.05
λ_t (temporal, $\mathcal{L}_{\text{temp}}$)	0.08
λ_ρ (ratio, $\mathcal{L}_{\text{ratio}}$)	0.05 (full); 0.0 elsewhere
λ_o (orthogonality)	0.01
λ_c (contrastive)	0.01
Inference	
Sampler	DDIM [35]
Steps	25
CFG scale s	$\{3.0, 5.0\}$ (reported per-experiment)

Effective-batch-size audit. The default B_{eff}=8 in Table 8 refers to the real-data design-study configuration used for the main ablations and the B_{eff} sweep in Table 7. Table 9 reports the effective batch sizes used by the newer long-run suites.

Table 9: Effective batch sizes used across run families.

Run family	Per-device	Accum.	$\mathbf{B}_{\text{eff}}$
Real design study default / 50 epoch	2	4	8
Real B_{eff} sweep (A5)	varies	varies	$\{2, 4, 8, 16\}$
Real full-scale 100-epoch suite	4	3	12
Real LOSO 100-epoch suite	4	3	12
Synthetic (S) loss sweep / LOSO 100-epoch suite	4	4	16

A4.2 Auxiliary loss definitions

Orthogonality regularizer. $\mathcal{L}_{\text{orth}}$ pulls the two independent physics tokens $\mathbf{T}_{\text{rpm}}, \mathbf{T}_{S_C}$ toward orthogonality so that the RPM and slip axes are encoded in separable subspaces:

$$\mathcal{L}_{\text{orth}} \;=\; \big|\cos(\mathbf{T}_{\text{rpm}}, \mathbf{T}_{S_C})\big|. \tag{25}$$

Soft contrastive regularizer. $\mathcal{L}_{\text{contr}}$ pulls RPM tokens of clips with similar V_{eff} closer in embedding space:

$$\mathcal{L}_{\text{contr}} \;=\; \frac{1}{B(B-1)} \sum_{i \neq j} w_{ij} \, \|\mathbf{T}_{\text{rpm}}^{(i)} - \mathbf{T}_{\text{rpm}}^{(j)}\|_2^2, \qquad w_{ij} \;=\; \exp\!\big(-|V_{\text{eff}}^{(i)} - V_{\text{eff}}^{(j)}|\big). \tag{26}$$

A4.3 Compute budget

Table 10 summarizes the compute resources used for training and inference.

Table 10: Compute budget and trainable parameter count.

Resource	Value
Training	
Hardware	Single RTX 4090 (24 GB)
GPU-hours (50-epoch)	$\approx$ 6–8 h
GPU-hours (100-epoch)	$\approx$ 12–16 h
Total (all ablations)	$\approx$ 500 GPU-hours
Peak GPU memory	$\approx$ 20 GB
Inference	
Resolution	496×48 (raw 2040×204, resize 0.25, aligned to 16×)
Frames	30
Latency per clip (DDIM 25-step)	$\approx$ 35–45 s
Throughput	$\approx$ 80–100 clips/GPU-h
Trainable parameters (LoRA + physics encoder)	
SVD-xt backbone (frozen)	1.5 B
LoRA (rank 32, all attn. layers)	$\approx$ 22 M
Physics encoder (MLPs + interaction)	$\approx$ 2 M
Total trainable	$\approx$ 24 M (1.6% of backbone)

A4.4 Latent–pixel correlation protocol

For the latent–pixel diagnostic of Sec. 5.3, we sample 120 real polishing clips total (30 at each of the four training RPMs) with T=14 frames, compute D^{pix} and D^{lat} per clip, and report Pearson r. Preprocessing mirrors the training path: full frame, resize factor 0.25, pixel values in $[0, 1]$, and direct VAE encoding without a $2x - 1$ pixel shift.

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